Linear Algebra Tutorials Week8: Vectors

Tutorial Session

Learning Goals

1. Vectors of real numbers
2. Norms of vectors and their properties
3. Inner products: interpretation and properties
4. Representing a vector as a linear combination
5. Vector spaces and independent sets
6. Vector equations of lines and planes

Task 1: Addition and Subtraction of Vectors

Context: LLM Word Embeddings (Semantic Similarity). In Large Language Models (LLMs) such as ChatGPT and search engines, words are converted into long lists of numbers (e.g., (0.2, -0.5, 0.8...)) called word embeddings so the computer can understand semantic similarity, such as "King" - "Man" + "Woman" $\approx$ "Queen".

The Problem

Given toy vectors for "King", "Man" and "Woman": $\mathbf{k} = (2, 3)$, $\mathbf{m} = (2, 2)$, and $\mathbf{w} = (-2, -2)$, respectively.

- Calculate word embedding $\mathbf{q}$ (for Queen) using the semantic formula.
- Write all vectors $\mathbf{k}, \mathbf{m}, \mathbf{w}, \mathbf{q}$ in **column notation**.
- Write all vectors in **unit vector notation** $(\hat{i}, \hat{j})$.
- Draw these vectors in the XY-plane.

Task 2: Length of Unit Vector

Compute the corresponding unit length vector $\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$ for:

1. $\mathbf{v}_1 = (3, 4)$
2. $\mathbf{v}_2 = (-1, -2, 3)$
3. $\mathbf{v}_3 = (-7, 2, -4, \sqrt{12})$

Task 3: Inner Product & Alignment

Part A: Compute the inner product $\mathbf{u} \cdot \mathbf{v}$ and the angle θ between:

- $(3, -2, 2)$ and $(1, 2, 2)$

Part B: For a machine learning algorithm, we need to use feature vectors of A, B and C. These feature vectors represent ratings of the users using real numbers in three dimensions, Dimension 1: risk-taking, Dimension 2: sociability and Dimension 3: novelty seeking. Feature vectors for users A, B, and C are as follows:

$$\mathbf{a} = (1, 0, 1), \quad \mathbf{b} = (2, 1, -2), \quad \mathbf{c} = \left(\frac{1}{2\sqrt{2}}, -\frac{\sqrt{3}}{2}, \frac{1}{2\sqrt{2}} \right)$$

1. Compute all pairwise inner products ($\mathbf{a} \cdot \mathbf{b}$, $\mathbf{b} \cdot \mathbf{c}$, $\mathbf{a} \cdot \mathbf{c}$).
2. Identify which users are most aligned in their social and behavioural traits, using the above results.

Task 4: Cross Product (Robotics Application)

Problem: You are training a robotic arm. The robot needs to pick up a package. To grasp it successfully, the robot's gripper must approach the object perpendicular to its surface (along the "Surface Normal").Your 3D camera has identified a small patch on the package's surface. It has extracted two vectors, $\vec{u}$ and $\vec{v}$, that lie flat on this surface patch: one is $\vec{u} = (2, 1, 0)$ and the other is $\vec{v} = (1, 3, 0)$. Calculate the Surface Normal $\vec{n} = \vec{u} \times \vec{v}$ for a robotic gripper.

- Vector $\vec{u} = (2, 1, 0)$
- Vector $\vec{v} = (1, 3, 0)$

Questions

1. Calculate $\vec{n}$.
2. Is this surface facing **"up"** (ceiling) or **"down"** (floor)?
3. Explain your reasoning based on the z-component.

Task 5: Higher Dimensional Vectors

1. **Proof:** Show that $\mathbf{c} \cdot \mathbf{d} = (\mathbf{a} \cdot \mathbf{a}) - (\mathbf{b} \cdot \mathbf{b})$ for any $\mathbf{a}, \mathbf{b}$ if $\mathbf{c} = \mathbf{a} + \mathbf{b}$ and $\mathbf{d} = \mathbf{a} - \mathbf{b}$.

2. **Calculation:** Given $\mathbf{a} = (1, 2, 3, 4, 5)$ and $\mathbf{b} = (1, \sqrt{2}, \sqrt{3}, \sqrt{4}, \sqrt{5})$, find:
   - $\mathbf{c} = \mathbf{a} + \mathbf{b}$
   - $\mathbf{d} = \mathbf{a} - \mathbf{b}$
   - The dot product $\mathbf{c} \cdot \mathbf{d}$

Task 6: Vector Equation of a Line

1. **2D:** Find the vector equation of the line $y = -x + 1$.
2. **3D:** Find the vector equation of the line $x + 1 = y + 2 = z + 3$.
3. **Conversion:** Find the Cartesian equation of the line:

$$\mathbf{r} = \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix} + t \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$$

Task 7: Vector Equation of a Plane

1. Find the vector equation of the plane $x + y + z = 2$.

2. Find the Cartesian equation of the plane containing the origin $(0, 0, 0)$ and having a normal vector:

$$\mathbf{n} = \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$$

Task 8: Equivalence of Plane Equations

Step 1: Find the Cartesian equation of the plane $\vec{r}_1$:

$$\vec{r}_1 = \begin{pmatrix}1\\1\\1\end{pmatrix} + s\begin{pmatrix}2\\1\\-1\end{pmatrix} + t\begin{pmatrix}0\\1\\2\end{pmatrix}$$

Step 2: Show that the following vector equation represents the **same** plane:

$$\vec{r}_2 = \begin{pmatrix}1\\2\\3\end{pmatrix} + s\begin{pmatrix}6\\2\\-5\end{pmatrix} + t\begin{pmatrix}2\\-2\\-7\end{pmatrix}$$